Sina Abdipoor (思纳), Ph.D.

AI Engineer specializing in Optimization, Operations Research, and Machine Learning

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Summary

AI Engineer with a Ph.D. in Computer Science and 5 years of experience as an Operations Research consultant. I build optimization and machine learning models for real scheduling, allocation, and forecasting problems, and I explain the results clearly to management.

Experience

Sunway University, Lecturer, Department of Data Science and Artificial Intelligence — Dec 2025 – present
Leading an optimization research group that solves real-world scheduling problems, and teaching optimization, machine learning, and software engineering.

Universiti Putra Malaysia, Graduate Researcher & Tutor — Sep 2020 – Sep 2025
Built hybrid metaheuristics that solve large real-world university timetabling problems, and taught programming and machine learning.

Independent Consultant, Operations Research Engineer — Jan 2020 – Mar 2025
- Analyzed client workflows to find inefficiencies in scheduling and resource allocation.
- Designed, built, and deployed optimization and prediction models for real operations.
- Presented findings and recommendations to stakeholders and management.

Pars Six Sigma Excellence, Software Engineer — Jul 2017 – Jul 2019
- Designed, built, and maintained a CRM system for employees and customers in a small software team.
- Developed the Java backend with a variety of libraries, frameworks, and APIs, and designed the SQL database.

IKCO, Android Developer (Internship) — Jun 2016 – Sep 2016
Built an Android app that lets assembly line workers check parts stock and order parts from their station.

Skills

Operations Research: Combinatorial Optimization, Mixed Integer Programming, Stochastic Optimization, Constraint Programming, Metaheuristics, Linear and Mathematical Programming, Gurobi, Google OR-Tools

Data Science: Predictive Modeling, Supervised and Unsupervised Learning, Data Visualization, TensorFlow, Scikit-Learn, NumPy, Pandas, Matplotlib, Power BI, Tableau, Excel

Programming: Python, Java, SQL, MySQL, SQLite, XML, Git, GitHub, UML

Languages: English (Bilingual, IELTS 8), Mandarin (Intermediate), Malay (Basic)

Projects

ESA Space Optimisation Competition (SpOC 4) — 2026
Ranked 3rd globally in the Luna Tomato Advertising challenge of the European Space Agency's space logistics optimization competition at GECCO 2026, using permutation encodings and simulated annealing. Our solutions were published as two GECCO 2026 papers.

International Timetabling Competition 2019 — 2022
An implementation of the International Timetabling Competition 2019 benchmark, the latest benchmark for university course timetabling, built as the core of my Ph.D.

Java Framework for Genetic Algorithms — 2019
A Java framework to implement evolutionary algorithms, apply them to different problems, and compare their performance. Built during my master's research.

Auto Parts Finder 2016
An Android app used by assembly line workers at IKCO to check parts stock in real time and order parts, which cut idle time on the line.

Education

Ph.D. **Universiti Putra Malaysia** 2020 – 2026
Computer Science
Thesis: Optimality and Generality of Metaheuristics in University Course Timetabling. CGPA: 4.0/4.0

M.Sc. **K. N. Toosi University of Technology** 2016 – 2019
Computer Engineering, Artificial Intelligence and Robotics
Thesis: Analyzing Parameter Tuning of Evolutionary Algorithms in Optimization Problems

Selected Publications

An integration of discrete wavelet transform and time-series recurrent neural network to improve stock price prediction Aug 2026
Keong Kuon Lock, Razali Yaakob, **Sina Abdipoor**, Mingxuan Li
Applied Computing and Informatics
10.1108/ACI-08-2025-0335

Optimality Versus Generality: Performance Assessment of Meta-Heuristics in Educational Timetabling 2025
Sina Abdipoor, Razali Yaakob, Say Leng Goh, Salwani Abdullah, Hazlina Hamdan, Khairul Azhar Kasmiran
IEEE Access
10.1109/ACCESS.2025.3565522

Hybrid Graph Neural Networks with LSTM Attention Mechanism for Recommendation Systems in MOOCs Mar 2024
Best Paper Award
Hao Luo, **Sina Abdipoor**, Qing Li, Li-Tao Xu
2024 20th IEEE International Colloquium on Signal Processing & Its Applications (CSPA)
10.1109/CSPA60979.2024.10525319

Awards

Best Paper Award Aug 2026
19th International Conference on Advanced Computer Theory and Engineering (ICACTE)

National Finalist, 3 Minute Thesis (3MT) Competition Jun 2024
Universiti Kebangsaan Malaysia

Winner, 3 Minute Thesis (3MT) Competition Mar 2024
Universiti Putra Malaysia

Talks

Applied Operations Research in Driving Sustainability China
Invited talk, Baoshan University Aug 2026

Applied Operations Research in Driving Sustainability China
Guest talk, Hefei University Aug 2026

Applied Operations Research in Developing Sustainable Cities Malaysia
Organizer and speaker, Sunway Tech Club, Sunway University Aug 2026

Certificates

Chinese (Mandarin) Certificate May 2021
Peking University

Google Data Analytics Certificate Apr 2021
Google